4 bowlers, 2 wicket keepers and 5 other players can be selected in

$4C_4 \times 2C_2 \times 10C_5$ ways

$4C_4 \times 2C_2 \times 10C_5 = 252$ ways

By addition principle of counting.

Total number of ways

$= 960 + 840 + 420 + 252$

$= 2472$

Example 2.32

If $4(nC_2) = (n+2)C_3$, find n

Solution

$$4(nC_2) = (n+2)C_3$$

$$4\frac{n(n-1)}{1\times 2} = \frac{(n+2)(n+1)(n)}{1\times 2\times 3}$$

$$12(n-1) = (n+2)\,(n+1)$$

$$12(n-1) = (n^2+3n+2)$$

$$n^2-9n+14 = 0$$

$$(n-2)(n-7) = 0 \Rightarrow n = 2, n = 7$$

Example 2.33

If $(n+2)C_n = 45$, find n

Solution

$$(n+2)C_n = 45$$

$$(n+2)C_{n+2-n} = 45$$

$$(n+2)C_2 = 45$$

$$\frac{(n+2)(n+1)}{2} = 45$$

$$n^2+3n-88 = 0$$

$$(n+11)(n-8) = 0$$

$$n = -11, 8$$

$n = -11$ is not possible

$$\therefore n = 8$$

Exercise 2.4

1. If $nP_r = 1680$ and $nC_r = 70$, find n and r.
2. Verify that $8C_4 + 8C_3 = 9C_4$.
3. How many chords can be drawn through 21 points on a circle?
4. How many triangles can be formed by joining the vertices of a hexagon?
5. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?
6. If four dice are rolled, find the number of possible outcomes in which atleast one die shows 2.
7. There are 18 guests at a dinner party. They have to sit 9 guests on either side of a long table, three particular persons decide to sit on one particular side and two others on the other side. In how many ways can the guests to be seated?
8. If a polygon has 44 diagonals, find the number of its sides.
9. In how many ways can a cricket team of 11 players be chosen out of a batch of 15 players?
   (i) There is no restriction on the selection.
   (ii) A particular player is always chosen.
   (iii) A particular player is never chosen.
10. A Committee of 5 is to be formed out of 6 gents and 4 ladies. In how many ways this can be done when
    (i) atleast two ladies are included
    (ii) atmost two ladies are included

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